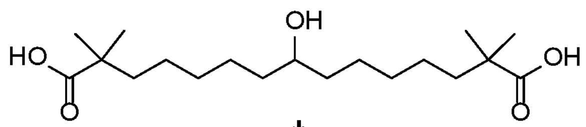
# PROCESS FOR PREPARATION OF BEMPEDOIC ACID AND ITS INTERMEDIATES

## FIELD OF INVENTION

The present application relates to a process for preparation of Bempedoic acid of formula (**I**) or its pharmaceutically acceptable salts thereof. The present application also relates to a novel intermediate (**II**) of Bempedoic acid, process for preparation thereof and use thereof in the preparation of Bempedoic aid.

## BACKGROUND ART

Bempedoic acid (**I**) is an ATP Citrate Lyase inhibitor that, reduces cholesterol biosynthesis and lowers LDL-C by up-regulating the LDL receptor. The drug compound having the adopted name “Bempedoic Acid” has chemical name 8-Hydroxy-2,2,14,14-tetramethylpentadecanedioic acid having the following structure:

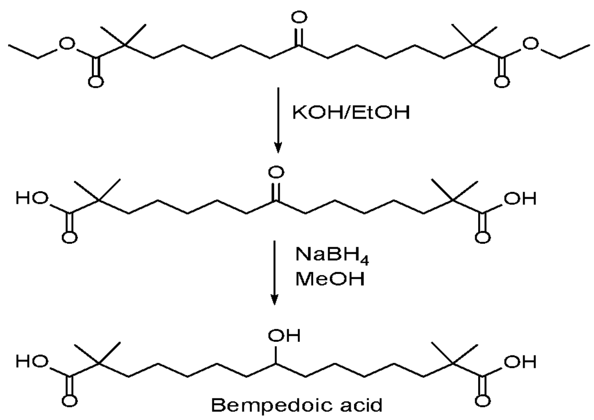


(I)

Both Bempedoic acid and a combination of Bempedoic acid/ Ezetimibe have been approved for the treatment of patients with elevated low-density lipoprotein cholesterol (LDL-C).

US7335799 discloses Bempedoic acid, its process of preparation or its pharmaceutically acceptable salt, hydrate, or solvate and pharmaceutical composition.

The following scheme-1 describe the process of Bempedoic in example 6.19 and 6.20 as given in US7335799.




**Scheme-1**

The process described above (Scheme-1) involves the use of excess amount (more than 4.2 mmol) of sodium borohydride ($NaBH_4$) to complete the reaction. In addition to this resulted Bempedoic acid is obtained in low yield (60%), as viscous oil, with low HPLC purity 83.8%. The low purity and viscous oily nature of Bempedoic acid is not suitable as active ingredient for use in pharmaceutical product.

Further, EP3666750A1, WO2020141419A2 and WO2020194152A1 disclose a process for preparation of Bempedoic acid by flipping the reduction and hydrolysis steps disclosed in US7335799 to provide solid Bempedoic acid with better purity. However, all those processes have limited use.

Still there is a need to provide a cost effective and commercially viable process for the preparation of highly pure Bempedoic acid which is suitable for use in pharmaceutical composition.

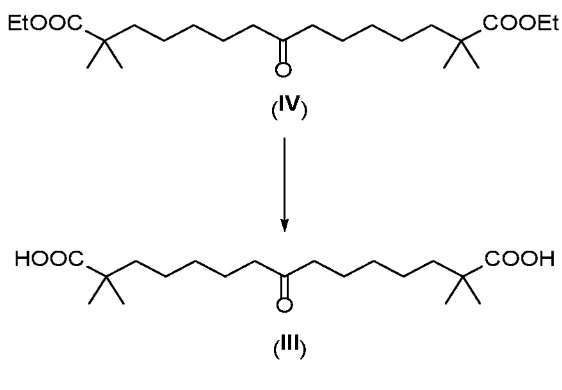
## BRIEF DESCRIPTION OF THE DRAWINGS

Figure 1 is an illustration of a PXRD pattern of a crystalline form of formula **(IIA)**, obtained by the Example-13.

## SUMMARY OF THE INVENTION

One aspect of the present application provides a process for preparation of compound of formula **(II)** comprising;

a) reacting compound of formula **(IV)** with suitable hydrolyzing agent to form compound of formula **(III)**

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b) converting the compound of Formula **(III)** to compound of formula **(II)**